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Thermodynamic modeling of the Si-Y system aided by first-principles and phonon calculations

Zhao Zhang[a], Mao-You Chu[a,*], Xu-Shan Zhao[b], Kun Li[a], Shun-Li Shang[c], Zi-Kui Liu[c], Jian-Yun Shen[a]

[a] Division of Rare Metals Metallurgy and Materials, General Research Institute for Nonferrous Metals, Beijing, 100088, PR China
[b] Computer Network Information Center, Chinese Academy of Sciences, Beijing, 100190, PR China
[c] Department of Materials Science and Engineering, The Pennsylvania State University, University Park, PA, 16802, USA

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ABSTRACT

Thermodynamic modeling of the Si-Y binary system has been performed by the CALPHAD (CALculation of PHAse Diagram) method based on phase diagram and thermochemical data in the literature combined with Gibbs energies of end-members of compounds predicted by first-principles phonon calculations. In particular, non-stoichiometric compounds Si_2Y and Si_3Y_5 are modelled to accommodate their homogeneity ranges in terms of two-sublattice models $(Si,Y)_2(Si,Y)$ and $(Si)_3(Si,Y)_5$, respectively. Formation of SiY is treated as a peritectic reaction according to experimental results, instead of an eutectic one as described in the previous models. The calculated phase equilibriums and thermodynamic properties are in a satisfactory agreement with available experimental data.

1. Introduction

Elements Si and Y are often simultaneously applied in many commercial and promising alloys. For example, the eutectic Si-Y alloy has potential applications in reactive melt infiltration to prepare the SiC_f/SiC composites [1]. MCrAlSiY (M = Ni, Co or Ni + Co) alloys are the bond coating (BC) materials for reducing thermal expansion mismatch between thermal barrier coatings (TBC) and matrix [2,3]. Addition of Si and Y are used to refine grains and improve the creep resistance in Mg-based alloys [4]. Si-Y co-deposition coatings improve oxidation resistance in Nb-based alloys [5,6]. An accurate thermodynamic modeling of the Si-Y system is thus important in understanding alloys containing both Si and Y.

Phase diagram and thermochemical properties of the Si-Y system have been extensively studied [7–15] and reviewed [16–19]. Thermodynamic models of Si-Y have been established [20–23]. This system was re-modelled in the present work due to the contradictions between existing modeling work [20–23] and experimental observations [7,8,17]. For example, the non-stoichiometric Si_2Y and Si_3Y_5 were treated as line compounds by Ran et al. [20], COST507 [21], Shukla et al. [22], and Kim and Jung [23]. The peritectic reaction, liquid + Si_4Y_5→SiY [7,8], was treated as an eutectic one by all of them [20–23].

In the present work, first-principles calculations based on the density functional theory (DFT) are applied to calculate thermodynamic properties of end-members in the sublattice models of Si_2Y and Si_3Y_5 in terms of the quasi-harmonic approximation (QHA) [24–26]. These first-principles results replace the very uncertain and often arbitrary values for some of them in the existing CALPHAD modeling. Thermodynamic parameters of the Si-Y system in the present work were evaluated using the PanOptimizer module in Pandat software [27].

2. Literature review

2.1. Phase diagram

Based on the X-ray diffraction (XRD) and metallographic results, Lundin [7] firstly constructed the Si-Y phase diagram with four compounds, SiY, Si_3Y_5, Si_4Y_5, and Si_5Y_3; three eutectic reactions (i) liquid→Si + Si_5Y_3 at 1215 °C and 82 at.% Si, (ii) liquid→Si_4Y_5 + Si_3Y_5 at 1710 °C and 41 at.% Si, and (iii) liquid→Si_3Y_5 + Y at 1260 °C and 13 at.% Si; and two peritectic reactions with SiY and Si_5Y_3 formed through at 1835 °C and 1635 °C, respectively. Note that the temperature of the reaction liquid→Si_4Y_5 + Si_3Y_5 is not very convincing with such a steep liquidus from 37.5 to 44.4 at.% Si at 1710 °C–1850 °C. Gschneidner [16] reported that the solid solubilities of Si in Y and Y in Si are less than

* Corresponding author.
E-mail address: chumaoyou@163.com (M.-Y. Chu).

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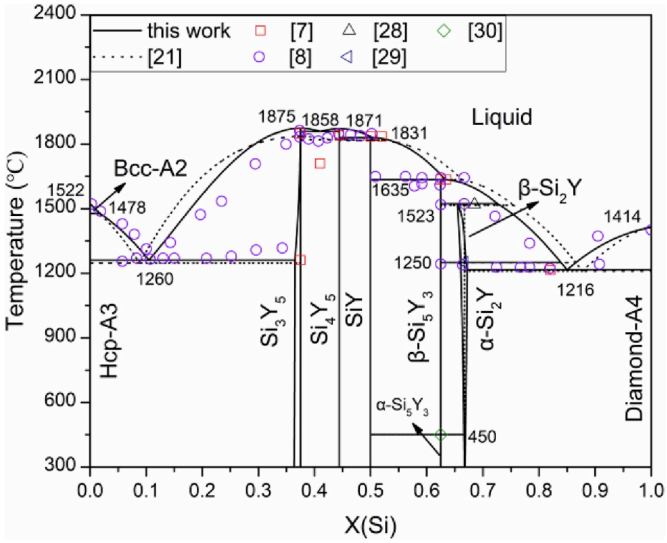

Fig. 1. Calculated phase diagram based on the present CALPHAD modeling (the solid lines) compared with the previous CALPHAD modeling (the dot lines) [21] and experiment data (the symbols) [7,8,28–30].

Table 1
Calculated temperature and compositions for invariant reactions by the present CALPHAD modeling together with experimental data.

Reaction	Composition (at.%Si)	Temperature (°C)	Reference
L→Si + α-Si_2Y	1/0.820/0.675	1215	[7]
	1/0.851/0.676	1216	This work
L + β-Si_5Y_3→β-Si_2Y	0.685/0.667/0.625	1520	[28]
	0.725/0.655/0.625	1523	This work
α-Si_2Y→β-Si_2Y	0.667	1250	[29]
	0.664/0.673	1250	This work
L + SiY→β-Si_5Y_3	0.635/0.635/0.500	1635	[7]
	0.661/0.635/0.500	1636	[8]
	0.633/0.625/0.500	1635	This work
α-Si_5Y_3→β-Si_5Y_3	0.625	450	[30]
	0.625	450	This work
L + Si_4Y_5→SiY	0.520/0.500/0.445	1835	[7]
	0.513/0.500/0.445	1836	[8]
	0.504/0.500/0.445	1831	This work
L→Si_4Y_5	0.444	1840	[7]
	0.444	1871	This work
L→Si_4Y_5 + Si_3Y_5	0.445/0.410/0.375	1710	[7]
	0.445/0.411/0.375	1858	This work
L→Si_3Y_5	0.375	1850	[7]
	0.375	1875	[8]
	0.375	1875	This work
L→Si_3Y_5 + Y	0.375/0.130/0	1260	[7]
	0.375/0.130/0	1260	[8]
	0.375/0.105/0	1260	This work

0.01 wt%. Binder [28] observed another binary compound Si_2Y formed peritectically at 1520 °C. Mayer et al. [29] suggested that Si_2Y has two polymorphs according to high temperature XRD results, i.e., low temperature α-Si_2Y with hexagonal structure (temperature T < 1250 °C) and high temperature β-Si_2Y with tetragonal structure (T > 1250 °C). Perri et al. [30] proposed that Si_5Y_3 undergoes a polymorphic transformation from orthorhombic (α-Si_3Y_5, T < 450 °C) to hexagonal structure (β-Si_3Y_5, T < 450 °C) according to Noreco diffractometer determination. Gokhale and Abbaschian [17] reviewed the Si-Y system and suggested that Si_2Y and Si_3Y_5 with the homogeneity ranges of 66–67.5 at.% Si and 36.5–37.5 at.% Si, respectively, according to the results of Dvorina and Verkhoglyadova [31].

Button et al. [8] reconstructed the Si-Y phase diagram according to the differential thermal analysis (DTA), metallography, XRD, and microhardness measurements. They suggested that the Si-Y system contains a peritectic reaction, liquid + Si_4Y_5→SiY, and the compound Si_3Y_5 with the highest melting point 1875 °C. They also pointed out that (i) the melting behavior of Si_4Y_5 is difficult to be determined because the samples of this composition always contain two phases (Si_3Y_5 + Si_4Y_5) and a very complex melting structure during heating and cooling; and (ii) Si_2Y cannot be formed, agreeing with Lundin's work [7]. The works of Meschel and Kleppa [14], Jiang et al. [32] and Li et al. [1] indicated that Si_2Y is stable.

All the experimental data of phase equilibrium and invariant reactions are shown in Fig. 1 and Table 1.

Table 2
Calculated enthalpies of formation ($\Delta H_{f,298}$) and Gibbs energies of formation ($\Delta G_{f,850}$) of the compounds by the present CALPHAD modeling (CALPHAD) and first-principles calculations (FP) with previous calculation and experimental data.

Phase	$-\Delta H_{f,298}$ (kJ/mol-atom)	Reference	$-\Delta G_{f,850}$ (kJ/mol-atom)	Reference
SiY	80.7 ± 1.3	[14]	89.5	[15]
	80.4	CALPHAD	79.4	CALPHAD
	80.4	FP	–	–
	82.5	[23]	79.8	[23]
	86.5	[22]	82.6	[22]
	80.3	[21]	71.8	[21]
	62.0	[20]	53.1	[20]
Si_4Y_5	–	–	95.6	[15]
	77.7	CALPHAD	77.3	CALPHAD
	77.7	FP	–	–
	80.4	[23]	78.3	[23]
	81.2	[22]	80.3	[22]
	77.6	[21]	69.9	[21]
	64.4	[20]	54.6	[20]
Si_3Y_5	80.4 ± 2.3	[14]	96.8	[15]
	71.7 ± 3.6	[13]	–	–
	71.1	CALPHAD	71.5	CALPHAD
	71.1	FP	–	–
	75.0	[23]	73.7	[23]
	76.3	[22]	75.7	[22]
	73.5	[21]	66.1	[21]
	60.7	[20]	55.2	[20]
α-Si_2Y	61.8 ± 1.4	[14]	63.6	[15]
	61.2	CALPHAD	62.8	CALPHAD
	51.3	FP	–	–
	60.3	[23]	58.6	[23]
	60.3	[22]	60.2	[22]
	73.7	[21]	64.3	[21]
	41.7	[20]	38.6	[20]

2.2. Thermochemical properties

Stukalo et al. [9] measured the enthalpies of mixing (ΔH_{mix}) of liquid phase at 2053 K in ranges of 0–30 at.% and 50 to 100 at.% Si by the high temperature calorimeter. Ryss et al. [10] measured the partial enthalpies of mixing ($\Delta \bar{H}_{mix}$) with 0–20 at.% and 60 to 100 at.% Si at 1870 K using the high temperature calorimeter, but only reported the smoothed curves. Esin et al. [11] measured $\Delta \bar{H}_{mix}$ with 60–100 at.% Si at 1880 K, which were consistent with the results of Ryss et al. [10]. Sudavtsova et al. [12] measured $\Delta \bar{H}_{mix}$ with 0–20 at.% Y at 1770 K by the high temperature isoperibol calorimeter. All these experimental ΔH_{mix} and $\Delta \bar{H}_{mix}$ are shown in Fig. 2(a) and (b), respectively.

Topor and Kleppa [13] measured the standard enthalpy of formation ($\Delta H_{f,298}$) of Si_3Y_5 with the solute-solvent drop calorimetry. Meschel and Kleppa [14] measured the standard enthalpies of formation of SiY, Si_2Y, and Si_3Y_5 using the high temperature direct synthesis calorimetry. The difference of their results for Si_3Y_5 is 8.7 kJ/mol-atom. Polotskaya et al. [15] measured the Gibbs energies of formation ($\Delta G_{f,850}$) for all the compounds at 850 K by the electromotive forces (EMF) method. Pecharsky et al. [33] determined the heat capacities of Si_4Y_5 at constant pressure from 0 to 200 K by the semi-adiabatic calorimetry. All these

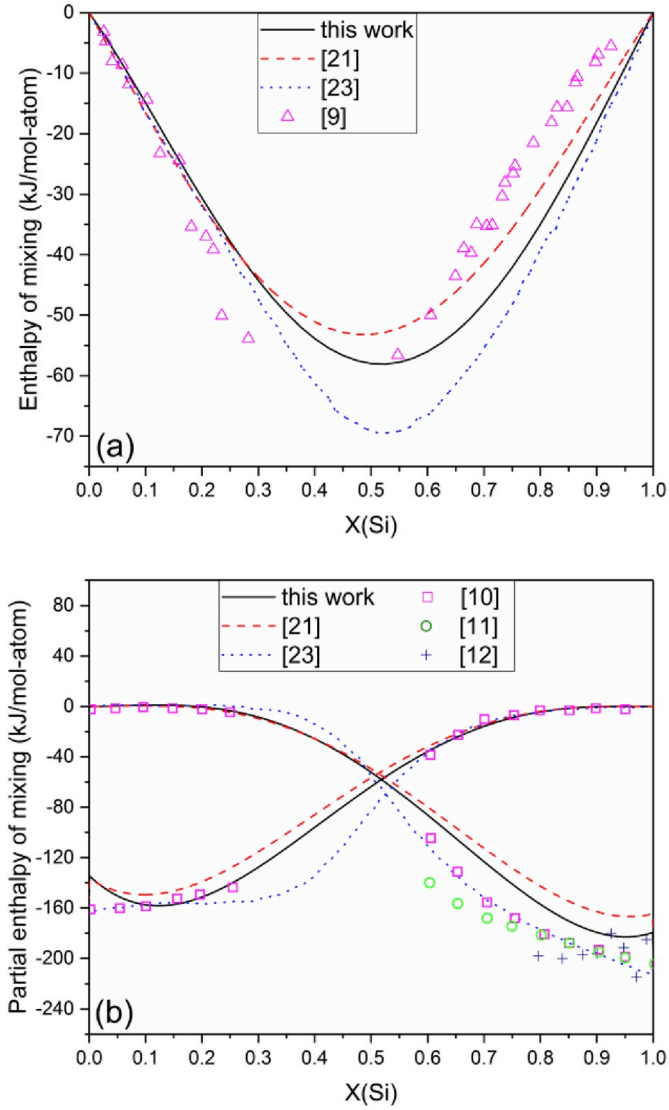

Fig. 2. (a) Enthalpies of mixing (ΔH_{mix}) and (b) partial enthalpies of mixing ($\Delta \bar{H}_{mix}$) of the liquid phase from the present CALPHAD modeling (the solid lines) with the previous calculations (the dash lines [21] and the dot lines [23]) and experiment data [9–12].

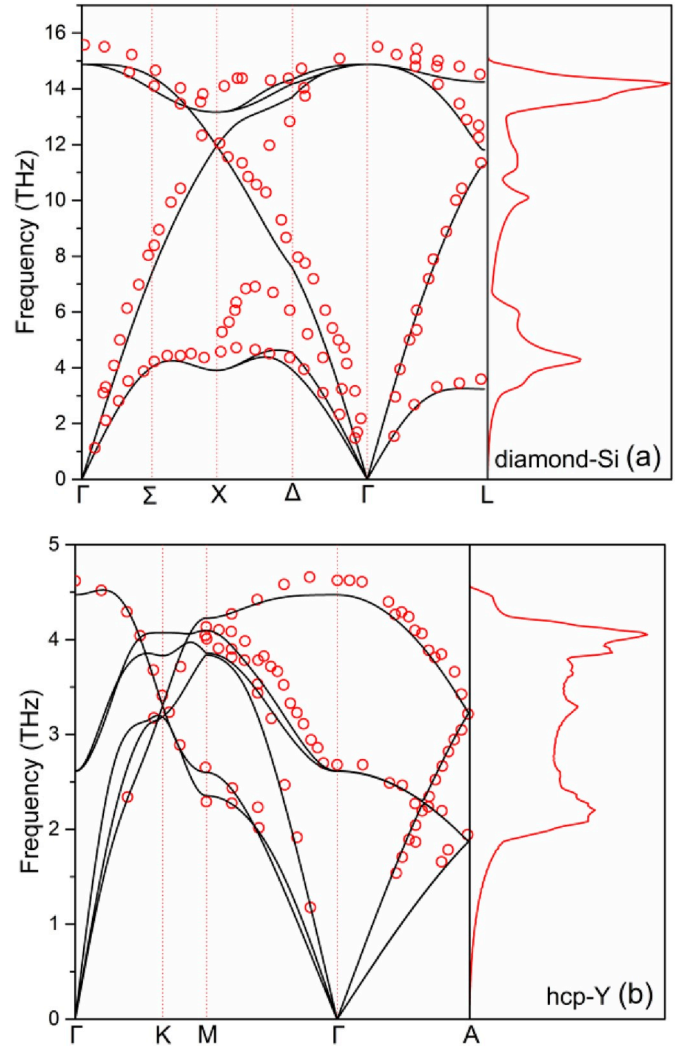

Fig. 3. Presently calculated (the lines) and the previously measured (the symbols) [48,49] phonon dispersion curves and density of states for diamond-Si and hcp-Y.

experimentally thermodynamic data are shown in Fig. 3.

3. Thermodynamic models

3.1. Pure elements

Gibbs energy for pure element i (i = Si and Y in present work) in phase ϕ is expressed by:

$$G_i^{0,\phi} = G_i^{\phi} - HSER = a + bT + cTlnT + dT^2 + eT^3 + fT^{-1} \quad (1)$$

where T is absolute temperature, and $HSER$ the molar enthalpy for the element of interest at 298.15 K and 1 bar in its SER (standard element reference) state. In the present work, the Gibbs energy functions of Si and Y were taken from the Scientific Group Thermodata Europe (SGTE) compiled by Dinsdale [34].

3.2. Solution phase

The Gibbs energies of solution phase such as liquid in the Si-Y system are modelled by the subregular solution model using the Redlich-Kister polynomial [35],

$$G_m^{\varphi} = x_{Si}\cdot G_{Si}^{\phi} + x_Y\cdot G_Y^{\phi} + R\cdot T\cdot(x_{Si}\cdot lnx_{Si} + x_Y\bullet lnx_Y) + x_{Si}\cdot x_Y\cdot[L_{Si,Y}^0 + L_{Si,Y}^1\cdot(x_{Si} - x_Y) + L_{Si,Y}^2\cdot(x_{Si} - x_Y)^2 + \cdots] \quad (2)$$

where x_{Si} and x_Y are the mole fractions of Si and Y in φ phase, R the gas constant, and L the binary interaction parameter expressed by,

$$L_{m,n}^{\nu} = A + B\cdot T \quad (3)$$

where A and B are parameters to be modelled.

3.3. Stoichiometric compounds

SiY, Si_5Y_3, and Si_4Y_5 are treated as stoichiometric compounds due to their narrow composition ranges. Gibbs energy of 1 mol Si_mY_n is expressed by,

$$G^{Si_mY_n} = m\cdot G_{Si}^{diamond} + n\cdot G_Y^{hcp} + A + B\cdot T \quad (4)$$

where A and B are parameters to be modelled.

3.4. Non-stoichiometric compounds

Si_2Y and Si_3Y_5 with homogeneity ranges of 32.5–34 at.% Y and 62.5–63.5 at.% Y [17], are modelled by the two-sublattice models,

Table 3
Crystallographic information and settings for DFT calculations of the Si-Y system.

Phase	Space group	Wyckoff symbol	k-mesh electron	Supercell size	k-mesh phonon	E_{cut} (eV)
Si [50]	Fd-3mS	8a-Si	12 × 12 × 12	2 × 2 × 2	5 × 5 × 5	600■ 370▲
Y [51]	P63/mmc	2c-Y	15 × 15 × 10	3 × 3 × 2	5 × 5 × 5	600■ 320▲
Si_4Y_5 [32]	Pnma	1d-Y 2c-Y 1d-Si 2c-Si	7 × 4 × 7	2 × 1 × 2	3 × 4 × 3	600■ 370▲
Si_3Y_5 [52]	P63/mcm	1g-Y 2d-Y 1g-Si	7 × 7 × 10	2 × 2 × 2	3 × 3 × 4	600■ 370▲
β-Si_2Y [28]	I41/amd	8e-Y 4a-Si	14 × 14 × 5	3 × 3 × 1	3 × 5 × 4	600■ 370▲
α-Si_2Y [28]	P6/mmm	1a-Y 1d-Si	15 × 15 × 15	3 × 3 × 3	3 × 5 × 4	600■ 370▲
SiY [53]	Cmcm	1c-Y 1c-Si	14 × 5 × 14	3 × 1 × 3	3 × 5 × 4	600■ 370▲

■ The cutoff energy used for structural relaxation of unit cell, ▲ The cutoff energy used for final calculations.

Table 4
Calculated structural properties of pure elements (diamond-Si and hcp-Y) and compounds of the Si-Y system including lattice parameters (*a*, *b*, and *c*), equilibrium volume (V_0), bulk modulus (B_0) and pressure derivative of bulk modulus (*B*').

Phase	Reference	Lattice parameters (Å)			E_0 (eV)	V_0 (Å^3)	*B* (GPa)	*B*'
		a	b	c				
Si	this work	5.43			−42.90	164.43	88.70	4.30
	Expt. [50,55,56]	5.46					97.83	4.24
Y	This work	3.66		5.73	−12.91	65.89	40.10	3.14
	Expt. [51,57]	3.65		5.69			41	3.15 (Calc, [58])
Si_4Y_5	This work	7.38	14.53	7.65	−243.67	840.98	80.98	3.89
	Expt. [32]	7.45	14.62	7.71				
Si_3Y_5	This work	8.40		6.30	−108.06	395.91	75.26	3.91
	Expt. [52]	8.46		6.39				
β-Si_2Y	This work	3.96		14.74	−75.26	233.21	85.60	4.52
	Expt. [28]	4.04		13.42				
α-Si_2Y	This work	3.85		4.14	−42.90	164.43	84.32	4.50
	Expt. [28]	4.09		3.95				
SiY	This work	4.25	10.5	3.82	−54.30	174.44	91.39	4.01
	Expt. [53]	4.28	10.6	3.85				

$(Si,Y)_2(Si,Y)_1$ and $(Si)_3(Si,Y)_5$, respectively. Gibbs energy of per mole Si_mY_n is expressed by,

$$G_m^{Si_mY_n} = y_{Si}^{'}y_{Si}^{''}G_{Si:Si}^{Si_mY_n} + y_Y^{'}y_{Si}^{''}G_{Y:Si}^{Si_mY_n} + y_{Si}^{'}y_Y^{''}G_{Si:Y}^{Si_mY_n} + y_Y^{'}y_Y^{''}G_{Y:Y}^{Si_mY_n} + mRT(y_{Si}^{'}\ln y_{Si}^{'} + y_Y^{'}\ln y_Y^{'}) + nRT(y_{Si}^{''}\ln y_{Si}^{''} + y_Y^{''}\ln y_Y^{''}) + G_m^{ex} \quad (5)$$

where $y_{Si}^{'}$ and $y_Y^{'}$ are the site fractions of Si and Y in the first sublattice, $y_{Si}^{''}$ and $y_Y^{''}$ are the site fractions in second sublattice, respectively. $G_{Si:Si}^{Si_mY_n}$, $G_{Y:Si}^{Si_mY_n}$, $G_{Si:Y}^{Si_mY_n}$, and $G_{Y:Y}^{Si_mY_n}$ represent the Gibbs energies of the end-members Si_mSi_n, Y_mSi_n, Si_mY_n, and Y_mY_n, respectively. G_m^{ex} is the excess Gibbs energy,

$$G_m^{ex} = y_{Si}^{'}y_Y^{'}\left(y_{Si}^{''}\cdot\sum_i L_{Si,Y:Si}^{i}(y_{Si}^{'}-y_Y^{'})^i + y_Y^{''}\cdot\sum_i L_{Si,Y:Y}^{i}(y_{Si}^{'}-y_Y^{'})^i\right) + y_{Si}^{''}y_Y^{''}\left(y_{Si}^{'}\cdot\sum_i L_{Si:Si,Y}^{i}(y_{Si}^{''}-y_Y^{''})^i + y_Y^{'}\cdot\sum_i L_{Y:Si,Y}^{i}(y_{Si}^{''}-y_Y^{''})^i\right) \quad (6)$$

where $L_{Si,Y:Si}^{i}$ represents *i*th interaction parameter between Si and Y in first sublattice when the second sublattice is fully occupied by Si. Correspondingly, $L_{Si:Si,Y}^{i}$ is *i*th interaction parameter in the second sublattice. Both $L_{Si,Y:Si}^{i}$ and $L_{Si:Si,Y}^{i}$ are expressed by Eq. (3).

4. First-principles methodology

According to the QHA, Gibbs energy, *G*, for condensed phase equals to Helmholtz energy, *F*, at pressure of 0 Pa [36,37], and can be expressed by,

$$G = F(V,T) + PV|_{p=0} = U + F_{vib}(V,T) + F_{el}(V,T) \quad (7)$$

where *V* is an equilibrium volume at the corresponding temperature.

In Eq. (7), *U* is the static energy at 0 K without contribution of zero-point vibrational energy and determined by fitting of the first-principles energy versus volume data points using a four-parameter Birch-Murnaghan equation of state (EOS) [38].

$$U(V) = a + bV^{-2/3} + cV^{-4/3} + dV^{-2} \quad (8)$$

where *a*, *b*, *c* and *d* are fitting parameters. The minimum value of EOS is the 0 K static energy at the equilibrium volume (V_0). Eleven fixed volume points with 0.01 × V_0 stepped from 0.95 × V_0 to 1.05 × V_0 were used for the present EOS fittings.

The second term, F_{vib}, accounts for the contribution of lattice vibrations to the Helmholtz energy. Once the phonon properties are obtained, thermodynamic properties at finite temperature from lattice vibrations can be calculated by statistical physics [39,40]. For example, the Helmholtz energy at the fixed volume and temperature is described by:

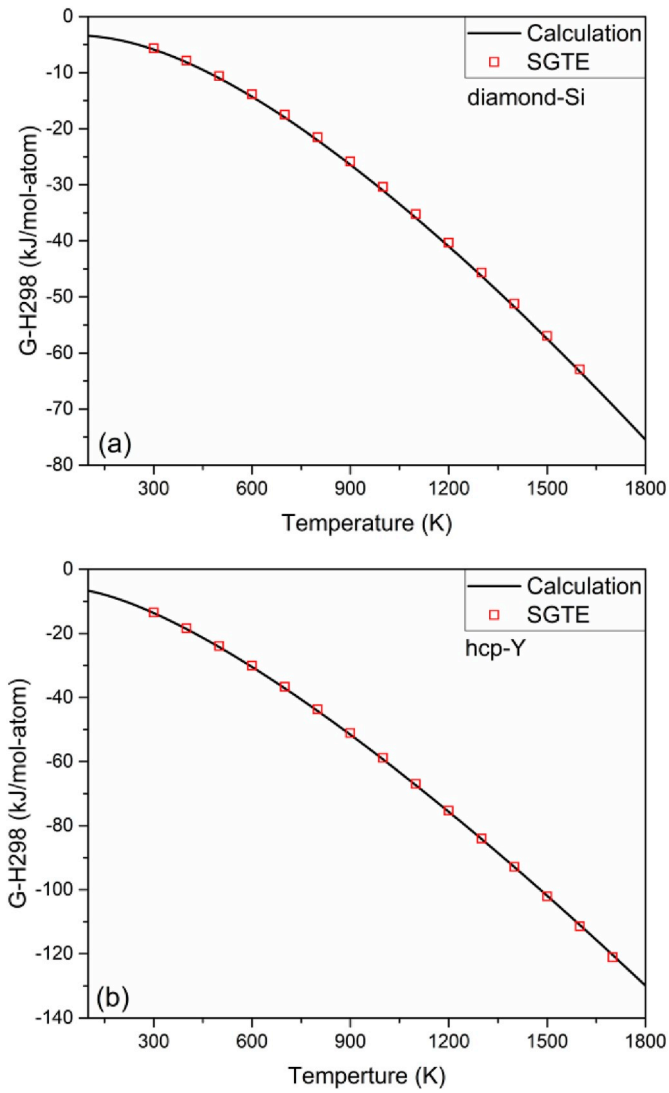

Fig. 4. Gibbs energies of pure diamond-Si (a) and hcp-Y (b) calculated by first-principles calculations (the lines, 100–1800 K)in comparison with the SGTE compilation [34] (the symbols, 300 K to melting points).

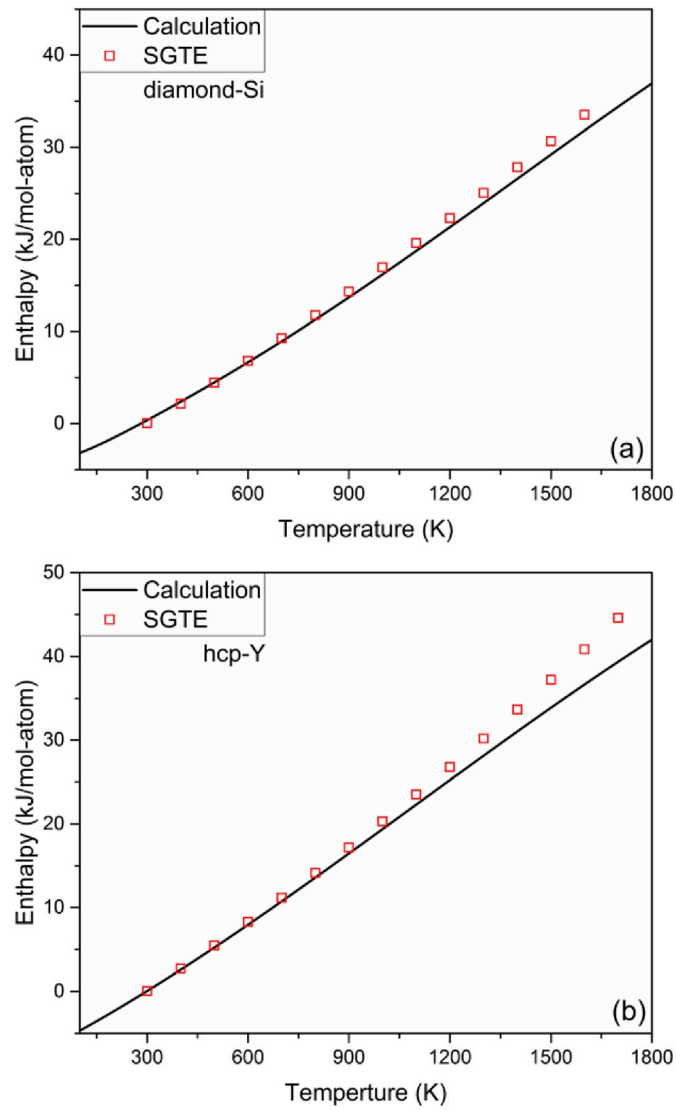

Fig. 5. Enthalpies of pure diamond-Si (a) and hcp-Y (b) calculated by first-principles calculations (the lines, 100–1800 K) in comparison with the SGTE compilation (the symbols, 300 K to melting points) [34].

$$F = -k_B T lnZ = U + \frac{1}{2}\sum_{qv} \hbar w(qv) + k_B T \sum_{qv} ln\left\{1 - \frac{exp[\hbar w(qv)]}{k_B T}\right\} \tag{9}$$

where k_B is the Boltzmann constant, Z the partition function, $\hbar$ the reduced Plank constant, and $w(qv)$ the frequency of phonon with wave vector q and polarization v.

The last part in Eq (7) F_{el}, represents the thermal electronic contribution to the Helmholtz energy, and can be described by Ref. [25],

$$F_{el} = E_{el} - T \cdot S_{el} \tag{10}$$

where E_{el} are S_{el} are the energy and entropy of thermal electron excitations, respectively.

In the present work, the Vienna *ab initio* Simulation Package (VASP) [41] and the PHONOPY package [42] were used to perform first-principles and phonon calculations. The electron-ion interactions were described by the projector augmented wave method (PAW) [43,44], and the generalized gradient approximation (GGA) by Perdew, Burke, and Ernzerhof (PBE) was used as the exchange-correlation functional [45]. The selected valance electronic configurations were $3s^2 3p^2$ and $4s^2 4p^6 4d^1 5s^2$ for Si and Y, respectively, as recommended by VASP. The energy convergence criterion of the electronic self-consistency was set to 10^{-6} eV/atom. To gain accurate energy and electronic structure, the Brillouin-zone integration is performed by tetrahedron method with Blöchl corrections [46].

For calculations of the vibrational energy, F_{vib}, the supercell approach [47] was carried out. Real-space force constants, i.e. the Hessian matrix, were calculated by the finite deference method as implemented in VASP. The phonon properties were calculated by PHONOPY. The energy cutoff (E_{cut}), k-points grids and simulation supercell sizes used for structural relaxations and phonon calculations are list in Table 3.

5. Result from first-principles calculations

Most end-members in sublattice models are not stable and their thermodynamic data are impossible to be determined by experiments. First-principles and phonon calculations are hence performed to estimate Gibbs energies of the compounds and end-members at finite temperature. To benchmark the calculation approach, the phonon dispersion curves and density of states (DOS) of diamond-Si and hcp-Y were calculated and compared with experimental data [48,49]. The thermodynamic properties of diamond-Si and hcp-Y were calculated

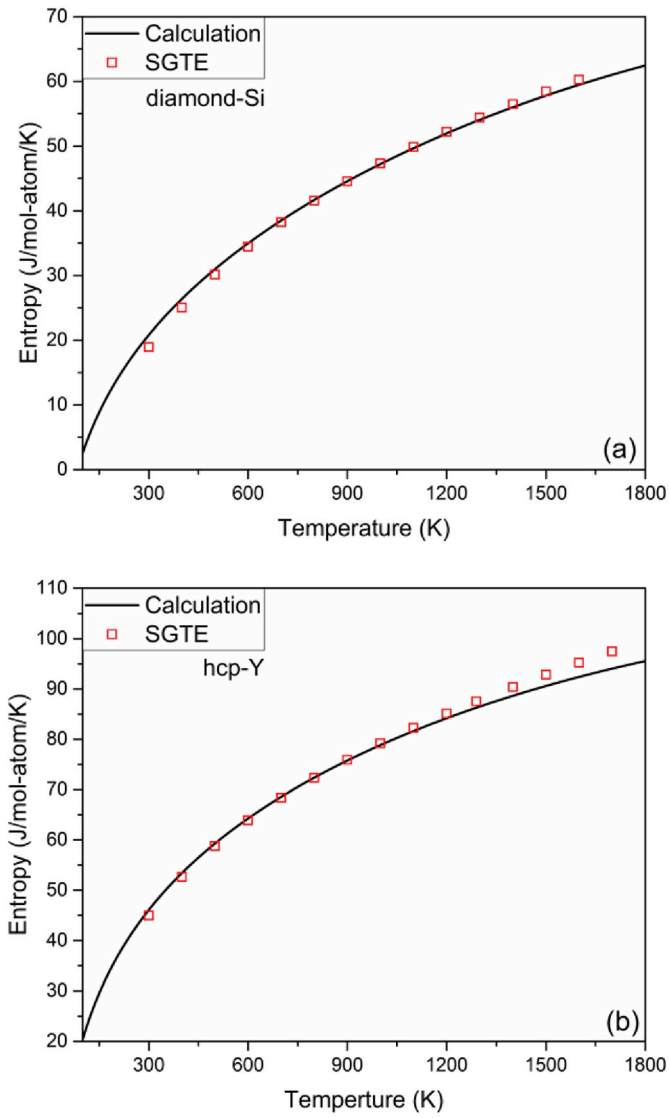

Fig. 6. Entropies of pure diamond-Si (a) and hcp-Y (b) calculated by first-principles calculations (the lines, 100–1800 K) in comparison with the SGTE compilation (the symbols, 300 K to melting points) [34].

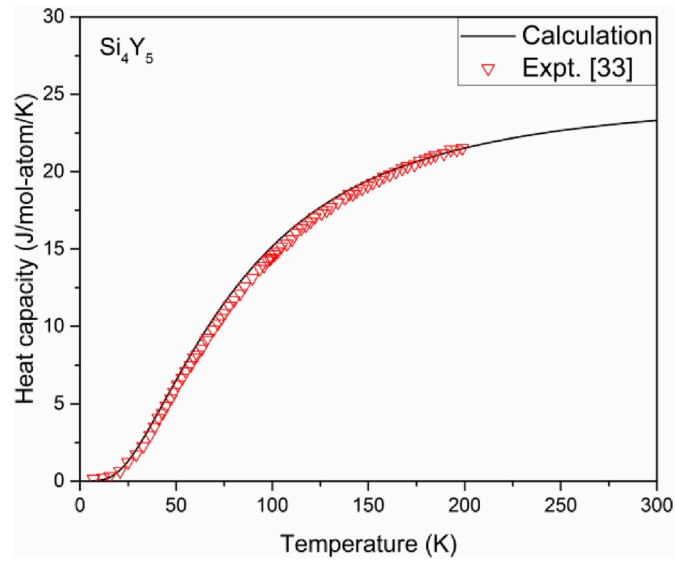

Fig. 7. Calculated heat capacity of Si_4Y_5 by first-principles phonon calculations (the lines, 0–300 K) with experimental data (the symbols, 0–200 K) [33].

and compared with experimental data compiled in the SGTE database [34]. The calculated heat capacity of Si_4Y_5 is compared with experimental data [33].

Table 3 lists the crystallographic information of diamond-Si [50], hcp-Y [51], α-Si_2Y [28], β-Si_2Y [28], Si_3Y_5 [52], Si_4Y_5 [32], and SiY [53]. Regarding Si_5Y_3 [30], it possesses orthorhombic structure (α-Si_5Y_3) with space group *Imma* at low temperatures and hexagonal structure (β-Si_5Y_3) with space group *P6/mmm* at high temperatures, respectively. In α-Si_5Y_3 and β-Si_5Y_3, Si in Wyckoff site 4e and 2d is not fully occupied, resulting in many configurations, and the first-principles calculations of their properties were thus not carried out. Table 4 shows the experimental and calculated equilibrium properties of the compounds and constitute elements derived from the EOS fittings using Eq. (8), including the E_0, V_0, bulk modulus (B_0), and pressure derivative of bulk modulus (B'). A good agreement is achieved between the calculated lattice parameters and the experimental data within 1.7% error except Si_2Y. It's likely caused by the neglection of van der Waals interaction between the layered structures in Si_2Y [54]. The calculated B_0 of diamond-Si is 9.13 Gpa smaller than experimental data of Jhon [55], while the value, 4.30 of B' agrees well with 4.24 by measurement [56]. As it can be seen in Table 4, both calculated B_0 and B' of hcp-Y are in a good agreement with experiments [57] and previous calculations [58].

The calculated phonon dispersion curves [48,49] of diamond-Si and hcp-Y along several high symmetry lines in the first Brillouin-zone are displayed in Fig. 3(a) and (b). It is shown that the calculated phonon frequencies of diamond-Si are just slightly different from measured result at 300 K in X → Γ, and the maximum difference with measured data is 3 THz. For hcp-Y, the calculation agrees well with experimental data and is only 0.2 THz smaller than measured result in M → Γ.

Thermodynamic properties of diamond-Si and hcp-Y are presented in Fig. 4 for the Gibbs energies, Fig. 5 for the enthalpies, and Fig. 6 for the entropies. It is seen from Figs. 4 and 5 that the calculated Gibbs energies and entropies by first-principles and phonon calculations from 100 to 1800 K agree very well with the SGTE data from 300 K to the melting points [34] within 3% error. The maximum deviation of enthalpies for hcp-Y in Fig. 5(b) is 5.2 kJ/mol-atom.

Comparison of calculated heat capacity of Si_4Y_5 with Pecharsky's results [33] is exhibited in Fig. 7. It can be seen a very small deviation within 0.5 J/mol-atom/K between the DFT calculations and experimental values at low temperatures (0–200 K).

All above results obtained by first-principles phonon calculations are in a good agreement with experimental data, indicating the present QHA can be applied to the Si-Y system. Fig. 8(a)–(e) present the phonon DOS of Si_4Y_5, Si_3Y_5, α-Si_2Y, β-Si_2Y and SiY. No imaginary modes are found, indicating that the compounds are dynamically stable at low temperatures. Table 2 lists the $\Delta H_{f,298}$ from first-principles calculations. It can be seen that all results agree well with experimental data [13,14] and the present CALPHAD modeling except α-Si_2Y with 10.3 kJ/mol-atom deviation. $\Delta H_{f,298}$ of SiY predicted by the current calculation is −80.4 kJ/mol-atom, which is in good consistency with the value, −80.7 kJ/mol-atom from the high temperature direct synthesis calorimetry [14]. Fig. 9 shows the vibrational entropies of compounds from 0 to 1800 K. It is found that vibrational entropies of β-Si_2Y exceeds that of α-Si_2Y at 1200 K, suggesting that more vibrational contribution to Gibbs energies of β-Si_2Y leads to the allotropic transformation of this phase.

6. Thermodynamic optimization and results

The evaluation of thermodynamic model parameters of the Si-Y system was carried out by Pandat software [27], while phase diagram and thermodynamic properties were calculated by Thermo-Calc

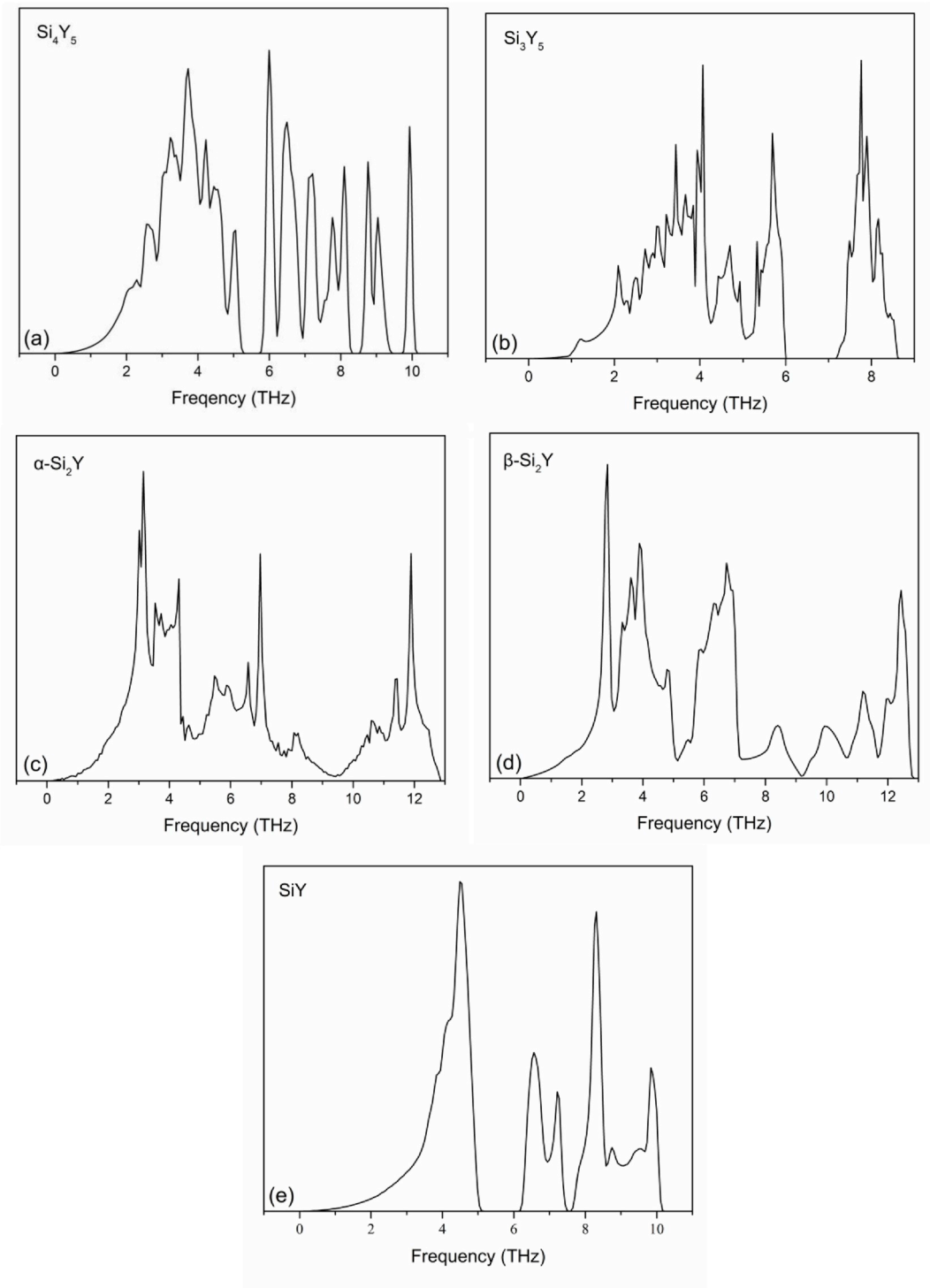

Fig. 8. Calculated phonon DOS of compounds in the Si-Y system: (a) Si_4Y_5; (b) Si_3Y_5; (c) α-Si_2Y; (d) β-Si_2Y; (e) SiY.

software [59]. Firstly, liquid interaction parameters L_0, L_1, and L_2 were modelled based on the experimental data of partial enthalpies and enthalpies of mixing [60]. Secondly, the Gibbs energies of end-members with off-stoichiometric compositions obtained by first-principles phonon calculations from 300 K to 2500 K were fitted using Eq. (4) and directly used in the sublattice models; while the parameters of other stable end-members were further adjusted according to the experimental data. The model parameters of α-Si_5Y_3 and β-Si_5Y_3 were totally evaluated based on phase equilibrium data. Finally, all the other parameters were optimized with the available experimental data. Table 5 lists the modelled thermodynamic parameters of Si-Y system.

Calculated phase diagram compared with experimental data [7,8,28–30] is shown in Fig. 1. The calculated liquidus is slightly different from the experimental data [8] in 20–40 at.% Si. Homogeneity ranges of Si_2Y and Si_3Y_5 in 66.4–67.3 at.% Si and 36.5 to 37.5 at.% Si, respectively are successfully reproduced. And as temperature decreases, the homogeneity range of Si_3Y_5 changes from 36.5 to 36.3 at.% Si below the temperature of the eutectic reaction, liquid → hcp-Y + Si_3Y_5.

Comparison of the calculated temperatures and compositions of invariant reactions with measured data is presented in Table 1. Most

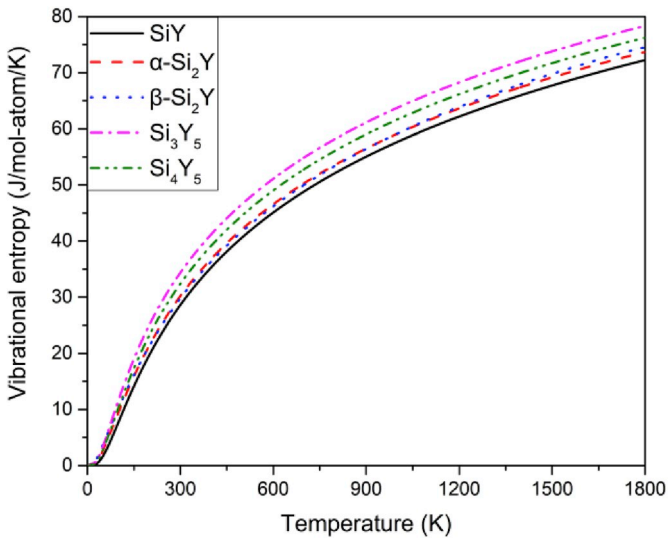

Fig. 9. Calculated vibrational entropies of compounds in the Si-Y system from first-principles phonon calculations at 0–1800 K.

calculated temperatures of invariant reactions agree well with experimental measurements. The peritectic reaction, liquid + Si_4Y_5→SiY, is well reproduced [7,8]. The eutectic temperature of hcp-Y and Si_3Y_5 is 1260 °C, which is in a good agreement with the experimental results of Lundin [7] and Button et al. [8].

Table 2 lists the calculated thermodynamic properties of the compounds in current CALPHAD modeling, first-principles calculations, previous CALPHAD modeling [21,23] and experimental data [13–15]. The calculated thermodynamic properties in this work are consistent with experimental data.

Fig. 2 presents the calculated ΔH_{mix}, and $\Delta \bar{H}_{mix}$ of liquid phase, compared with experimental data [9–12]. The calculated results by this modeling are more consistent with the experimental data than the previous modeling. The minimum value of the calculated mixed enthalpy is on the Si-rich side, which is different from the measured result by Stukalo et al. [9]. This difference may be caused by considering $\Delta \bar{H}_{mix}$ during optimization. In fact, the measured $\Delta \bar{H}_{mix}$ values [10–12] is inconsistent with experimental ΔH_{mix} values [9].

Fig. S1 shows the comparison of the calculated heat capacities for the Si-Y compounds by first-principles calculations and CALPHAD modeling. As shown in Fig. S1, an overall agreement is reached except for Si_3Y_5, α-Si_2Y, and SiY. Differences of the calculated heat capacities for Si_4Y_5 and β-Si_2Y by these two methods are within 5 J/mol-atom/K.

The entropy of the Si-Y compounds from first-principles calculations and CALPHAD modeling is displayed in Fig. S2. This figure shows a good agreement is achieved. The errors between these two methods are less than 10 J/mol-atom/K even at high temperatures.

7. Conclusions

Thermodynamic modeling of the Si-Y system has been re-established. The problems concerning peritectic reaction, liquid + Si_4Y_5 → SiY, and homogeneity ranges of Si_3Y_5 and Si_2Y in previous modeling are remedied. First-principles phonon calculations with QHA approach were performed to calculate the thermodynamic properties at finite temperature, the phonon dispersion curves and DOS for pure elements (diamond-Si and hcp-Y), agreeing well with measurements. Thermodynamic properties, phonon DOS of Si_4Y_5, SiY and Si_3Y_5, Si_2Y in terms of two-sublattice model $(Si,Y)_3(Y)_5$ and $(Si,Y)_2(Si,Y)$ were predicted to make up the shortage of experimental data. A set of self-consistent thermodynamic parameters is obtained by the CALPHAD approach. The present work provides a more accurate modeling of the Si-Y system for multicomponent system containing both Si and Y.

Table 5
Thermodynamic parameters in the Si-Y system. Gibbs energy is given in J/mol-atom, temperature (*T*) in Kelvin.

Phase	Parameters
Liquid:(Si,Y)	$L^0_{Si,Y}$ = -231878.58
	$L^1_{Si,Y}$ = -22570.43 + 3.00*T
	$L^2_{Si,Y}$ = 75072.9-26.56*T
α-Si_2Y:(Si,Y)$_2$:(Si,Y)	$G^{\alpha-Si_2Y}$ = 2*$G^{diamond}_{Si}$ + G^{hcp}_{Y}-183889.00-5.26*T
	$G^{\alpha-Si_2Si}$ = 3* $G^{diamond}_{Si}$ + 61229.34 + 25.79*T
	$G^{\alpha-Y_2Si}$ = 2*G^{hcp}_{Y} + $G^{diamond}_{Si}$ + 138760.97 + 5.55*T
	$G^{\alpha-Y_2Y}$ = 3* G^{hcp}_{Y} + 217959.61 + 36.01*T
	$L^0_{Si,Y:*}$ = -615442.00
	$L^0_{*:Si,Y}$ = -49575.22
β-Si_2Y:(Si,Y)$_2$:(Si,Y)	$G^{\beta-Si_2Y}$ = 2*$G^{diamond}_{Si}$ + G^{hcp}_{Y}-181158.00-7.05*T
	$G^{\beta-Si_2Si}$ = 3* $G^{diamond}_{Si}$ + 60907.00 + 24.72*T
	$G^{\beta-Y_2Si}$ = 2*G^{hcp}_{Y} + $G^{diamond}_{Si}$ + 134777.42 + 5.36*T
	$G^{\beta-Y_2Y}$ = 3* G^{hcp}_{Y} + 205090.07 + 25.90*T
	$L^0_{Si,Y:*}$ = -585927.00
	$L^0_{*:Si,Y}$ = -47560.00
Si_3Y_5:(Si,Y)$_{0.375}$:(Y)$_{0.625}$	$G^{Si_{0.375}Y_{0.625}}$ = 0.375* $G^{diamond}_{Si}$ + 0.625*G^{hcp}_{Y}-73054.19 + 0.41*T
	$G^{Y_{0.375}Y_{0.625}}$ = G^{hcp}_{Y} + 58325.09 + 9.13*T
	$L^0_{Si,Y:Y}$ = -62760.32 + 3.33*T
Si_4Y_5:(Si)$_4$:(Y)$_5$	$G^{Si_4Y_5}$ = 4* $G^{diamond}_{Si}$ + 5*G^{hcp}_{Y}-698969.80 + 3.19*T
SiY:(Si):(Y)	G^{SiY} = $G^{diamond}_{Si}$ + G^{hcp}_{Y}-160861.72 + 2.52*T
α-Si_5Y_3:(Si)$_5$:(Y)$_3$	$G^{\alpha-Si_5Y_3}$ = 5* $G^{diamond}_{Si}$ + 3*G^{hcp}_{Y}-557272.00 + 16.73*T
β-Si_5Y_3:(Si)$_5$:(Y)$_3$	$G^{\beta-Si_5Y_3}$ = 5* $G^{diamond}_{Si}$ + 3*G^{hcp}_{Y}-535920.00-12.78*T

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2019.03.009.

References

[1] J.G. Li, R. Hu, J.R. Yang, Y.S. Liu, Competitive growth of Si and YSi_2 phases in a eutectic Si-Y alloy prepared by the Bridgeman method, Ceram. Int. 44 (2018) 13232–13239.
[2] S.T. Zhang, K.P. Du, X.J. Ren, J. Shi, Effect of Si on hot corrosion resistance of CoCrAlY coating, Rare Metal Mater. Eng. 46 (2017) 2807–2811.
[3] H. Wang, Q.S. Wang, Research progress on modification in compositions of MCrAlY coatings, Surf. Technol. 43 (2014) 142–157.
[4] D.J. Zhao, X.F. Guo, W.P. Yang, C.J. Xu, B. Hu, X. Gao, Preparation and microstructure analysis of Mg-Y-Si alloy, Ordnance Material Sci. Eng. 34 (2011) 23–27.
[5] B.P. Bewlay, M.R. Jackson, P.R. Subramanian, J.C. Zhao, A review of very-high-temperature Nb-silicide-based composites, Metall. Mater. Trans. 34 (2003) 2043–2052.
[6] X.D. Tian, X.P. Guo, Structure and oxidation behavior of Si–Y co-deposition coatings on an Nb silicides, Surf. Coating. Technol. 204 (2009) 313–318.
[7] Lundin, Rare Earth Metal Phase Diagrams, J.Wiley and Sons, New York, 1961, pp. 264–266.
[8] T.W. Button, I.J. Mcolm, J.M. Ward, Preparation of yttrium silicides and oxide-silicides, J. Less Common. Met. 159 (1990) 205–222.
[9] V.A. Stukalo, G.I. Batalin, N.Y. Neshchimenko, V.P. Kurach, Formation enthalpy of molten alloys of yttrium with silicon, Ukr. Khim. Zh. 46 (1980) 98–99.
[10] G.M. Ryss, Y.O. Esin, M.S. Petrushevskii, Formation enthalpies of melts of yttrium with silicon, Izv. Akad. Nauk. SSSR, Met. 6 (1979) 67–70.
[11] Y.O. Esin, M.G. Valiishev, P.V. Geld, L.I. Tushkova, Partial and integral formation enthalpies of molten alloys of yttrium with silicon, Izv. Akad. Nauk. SSSR, Met. 1 (1976) 19–20.
[12] V.S. Sudavtsova, N.V. Kotova, Thermodynamic properties of Si-Y and Ge-Y melts, Inorg. Mater. 43 (2007) 647–652.
[13] L. Topor, O.J. Kleppa, Standard enthalpies of formation of Me_5Si_3 (Me = Y, Lu, Zr) and Hf_3Si_2, J. Less Common. Met. 167 (1990) 91–99.
[14] S.V. Meschel, O.J. Kleppa, Standard Enthalpies of Formation of some 4d transition metal silicides by high temperature direct synthesis calorimetry, J. Alloy. Comp. 274 (1998) 193–200.
[15] R.I. Polotskaya, V.R. Sidorko, Thermodynamic properties of yttrium silicides, Powder Metall. Met Ceram. 36 (1997) 315–319.
[16] K.A. Gschneidner, Yttrium-silicon System, Rare Earth Alloys, D. Van Nostrand Co., Inc., New York, 1961, pp. 276–278.
[17] A.B. Gokhale, G.J. Abbaschian, The Si-Y(Silicon-Yttrium) system, Bull. Alloy Phase Diagrams 5 (1986) 485–489.
[18] H. Okamoto, Si-Y (silicon-yttrium), J. Phase Equilibria Diffusion 12 (1991) 397–399.
[19] H. Okamoto, Si-Y (silicon-yttrium), J. Phase Equilibria Diffusion 32 (2011) 475–476.
[20] Q.S. Ran, H.L. Lukas, G. Effenberg, G. Petzow, A thermodynamic assessment of the Y-Si system, Z. Metallkd 80 (1989) 402–405.
[21] I. Ansara, A.T. Dinsdale, M.H. Rand, COST 507: Definition of Thermochemical and Thermophysical Properties to Provide a Database for the Development of New Light Alloys-Thermochemical Database for Light Metal Alloys, Office for Official Publications of the European Communities, Luxembourg, 1998, pp. 274–277.
[22] A. Shukla, Y.B. Kang, A.D. Pelton, Thermodynamic assessment of the Ce-Si, Y-Si, Mg-Ce-Si and Mg-Y-Si system, Int. J. Mater. Res. 100 (2009) 208–217.
[23] J.H. Kim, I.H. Jung, Critical evaluation and thermodynamic optimisation of the Si-RE system: Part II. Si-RE system (RE = Gd, Tb, Dy, Ho, Er, Tm, Lu and Y), J. Chem. Thermodyn. 81 (2015) 273–297.
[24] X.S. Zhao, G.H. Yuan, M.Y. Yao, Q. Yue, J.Y. Shen, First-principles calculations and thermodynamic modeling of the V-Zr system, Calphad 36 (2011) 163–168.
[25] Y. Wang, Z.K. Liu, L.Q. Chen, Thermodynamic properties calculation of Al, Ni, NiAl and Ni3Al from first-principles calculations, Acta Mater. 52 (2004) 2665–2671.
[26] S.L. Shang, Y. Wang, Z.K. Liu, First-principles calculations of phonon and thermodynamic properties in the boron-alkaline earth metal binary systems: B-Ca, B-Sr and B-Ba, Phys. Rev. B 75 (2007) 024302.
[27] W. Cao, S.L. Chen, F. Zhang, K. Wu, Y. Yang, Y.A. Chang, R. Schmid-Fetzer, W.A. Oates, PANDAT software with PanEngine, PanOptimizer and PanPrecipitation for multi-component phase diagram calculation and materials property simulation, Calphad 33 (2009) 328–342.
[28] I. Binder, Structure and properties of rare-earth and yttrium hard metals, J. Am. Ceram. Soc. 43 (1960) 287–292.
[29] I.P. Mayer, E. Banks, B. Post, Rare earth disilicides, J. Phys. Chem. 66 (1962) 693–696.
[30] J.A. Perri, E. Banks, B. Post, Polymorphism of rare earth disilicides, J. Phys. Chem. 63 (1959) 2073–2074.
[31] L.A. Dvorina, T.S. Verkhoglyadova, Production of rare earth metals silicides by arc melting, Russ. Metall. 6 (1965) 38–44.
[32] W.P. Jiang, Y.Z. Zhan, C.L. Li, S. Liu, H.X. Zhou, Y.H. Zhuang, Phase equilibria in Nb-Y-Si ternary system at 873K, J. Rare Earths 30 (2012) 934–940.
[33] A.O. Pecharsky, V.K. Pecharsky, K.A. Gschneidner, Phase relationships and low temperature heat capacities of alloys in the Y_5Si_4–Y_5Ge_4 pseudo binary system, J. Alloy. Comp. 379 (2004) 127–134.
[34] A.T. Dinsdale, SGTE data for pure elements, Calphad 15 (1991) 317–425.
[35] O. Redlich, A.T. Kister, Algebraic representation of thermodynamic properties and the classification of solution, Ind. Eng. Chem. 40 (1948) 345–348.
[36] Z.K. Liu, First-principles calculation and CALPHAD modeling of thermodynamics, J. Phase Equilibria Diffusion 30 (2009) 517–534.
[37] Y. Chen, Ph.D's Thesis, General Research Institute for Nonferrous Metals, Beijing, 2012.
[38] S.L. Shang, Y. Wang, D.E. Kim, Z.K. Liu, First-principles thermodynamics from phonon and Debye model: Application to Ni and Ni3Al, Comput. Mater. Sci. 47 (2010) 1040–1048.
[39] Y. Wang, S.L. Shang, H.Z. Fang, Z.K. Liu, L.Q. Chen, First-principles calculations of lattice dynamics and thermal properties of polar solids, NPJ Comput. Mater. 2 (2016) 16006.
[40] R. Wang, S.F. Wang, X.Z. Wu, A.P. Liu, First-principles phonon calculations of thermodynamic properties for ductile rare-earth intermetallic compounds, Intermetallics 19 (2011) 1599–1604.
[41] G. Kressea, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set, Comput. Mater. Sci. 6 (1996) 15–50.
[42] A. Togo, I. Tanaka, First principles phonon calculations in materials science, Scripta Mater. 108 (2015) 1–5.
[43] P.E. Blöchl, Projector augmented-wave method, Phys. Rev. B 9 (1994) 17953–17979.
[44] G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, Phys. Rev. B 54 (1996) 11169–11186.
[45] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized gradient approximation made simple, Phys. Rev. Lett. 77 (1996) 3865–3868.
[46] P.E. Blöchl, Improved tetrahedron method for Brillouin-zone integrations, Phys. Rev. B 49 (1994) 16223–16234.
[47] A. van de Walle, G. Ceder, The effect of lattice vibrations on substitutional alloy thermodynamics, Rev. Mod. Phys. 74 (2002) 11–45.
[48] G. Nilsson, G. Nelin, Study of homology between silicon and germanium by thermal-neutron spectrometry, Phys. Rev. B 6 (1972) 3777–3786.
[49] S.K. Sinha, T.O. Brun, L.D. Muhlestein, J. Sakurai, Lattice dynamics of yttrium at 295 K, Phys. Rev. B 1 (1970) 2430–2441.
[50] W.L. Bond, W. Kaiser, Interstitial versus substitutional oxygen in silicon, J. Phys. Chem. Solids 16 (1960) 44–45.
[51] F.H. Spedding, A.H. Daane, K.W. Herrmann, The crystal structures and lattice parameters of high-purity scandium, yttrium, and the rare earth metals, Acta Crystallogr. 9 (1956) 559–563.
[52] B.Y. Kotur, I.R. Mokra, Ternary systems Sc-Y-Si and Sc-Lu-Si, Inorg. Mater. 30 (1994) 729–732.
[53] V.I. Lazorenko, B.M. Rud, Y.N. Paderno, Some physical properties of yttrium silicides, Inorg. Mater. 10 (1974) 987–988.
[54] Y.J. Hu, J.P.S. Palma, Y. Wang, S.A. Firdosy, K.E. Star, Thermodynamic modeling of the La-Te system aided by first-principles calculations, Calphad 61 (2018) 227–236.
[55] J.H. Jhon, Electronic effects in the elastic constants of n-type silicon, Phys. Rev. 161 (1967) 756-751.
[56] G. Misra, P. Tripathi, S.C. Goyal, Bulk modulus of semiconductors and its pressure derivatives, Phil. Mag. Lett. 87 (2007) 393–402.
[57] B.J. Min, K.M. Ho, First-principles total-energy calculations on Y and α-$YH_{0.5}$, Phys. Rev. B 40 (1989) 7532–7536.
[58] S.L. Shang, A. Saengdeejing, Z.G. Mei, D.E. Kim, H. Zhang, S. Ganeshan, Y. Wang, Z.K. Liu, First-principles calculations of pure elements: equations of state and elastic stiffness constants, Comput. Mater. Sci. 48 (2010) 813–826.
[59] B. Sundman, B. Jansson, J.O. Andersson, The Thermo-Calc databank system, Calphad 9 (1985) 153–190.
[60] H.L. Lukas, S. Fries, B. Sundman, Computational Thermodynamics:The CALPHAD Method, Cambridge university press, New York, 2007, pp. 169–172.